

Atmanirbhar by Design: Building the Supply Chains Behind India's Solar Leadership

India's rise to become the world's second-largest solar market, backed by a record addition of 44.6 GW in FY26 and cumulative installed capacity of 154.24 GW as of April 2026, marks a defining milestone in the country's clean energy journey.

This achievement reflects sustained policy commitment. Programmes such as PM-KUSUM, PM Surya Ghar and large-scale renewable auctions have accelerated deployment, while the PLI Scheme, ALMM and customs duty measures have catalysed domestic manufacturing growth.

But the next chapter will not be defined by deployment alone.

Every industry reaches a moment when the conversation it has been having must evolve into the conversation it should be having. Indian solar manufacturing is at that point today.

Manufacturing depth is not measured in modules alone. A solar panel is the final product of a layered value chain comprising polysilicon, ingots, wafers, cells and modules. India has built the final layer of this ecosystem at genuine scale. Module manufacturing capacity today stands at approximately 210 GW. Cell manufacturing capacity is around 27 GW. Wafer and ingot capacity, however, remains close to 2 GW, while commercial polysilicon manufacturing is yet to be established in the country.

That gap between 210 GW of module capacity and 2 GW of wafer capacity may well be the most important number in Indian solar manufacturing today. It is not a gap in ambition. It is a gap in supply-chain sovereignty.

For as long as critical upstream inputs remain concentrated outside India, domestic manufacturers will continue to face structural vulnerabilities that no amount of module capacity can fully offset. Volatility in polysilicon

prices directly impacts production economics. Disruptions in upstream supply chains create bottlenecks that affect manufacturing plans, investment decisions and long-term competitiveness. The energy security India is building through renewable energy deployment requires a corresponding security in manufacturing. That security resides upstream.

The industry has recognised this reality. India's push towards nearly 33 GW of wafer and ingot manufacturing capacity by FY2030, backed by more than ₹80,000 crore of planned investments, reflects a growing understanding that the next manufacturing frontier lies in cells, wafers and ingots, where value addition is deeper, technological barriers are higher and strategic importance is significantly greater.

At Avaada, this conviction has shaped our manufacturing journey. Through Avaada Electro, we have established 8.5 GW of operational solar module manufacturing capacity and are actively progressing towards deeper integration into cell manufacturing. The objective is not merely scale; it is resilience. Manufacturers that control their upstream inputs gain greater control over costs, reliability, quality and competitiveness. At a national level, the same principle applies. An industry built on domestically integrated supply chains is one that can genuinely claim to be Atmanirbhar.

Upstream manufacturing demands more in capital, in technology and in time. Yet these are precisely the reasons why the companies that build these capabilities today will shape Indian solar manufacturing for decades.

The global market compounds this opportunity. As countries seek diversified clean energy supply chains, India is well placed to become a trusted manufacturing hub. This opportunity is time-sensitive — the countries that build integrated

supply chains over the next decade will shape the future of global energy trade and technological leadership.

The first phase of India's solar journey was about adding gigawatts.

The next phase is about building capability across the entire value chain.

Because true Atmanirbharta is achieved not merely when a nation installs clean energy at scale, but when it designs, manufactures and innovates across every layer of the ecosystem that powers it.

India has every reason to lead that future.



Vineet Mittal
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